

Introduction to Logic I

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Abstract

Notes for the University of Helsinki course MAT21014, taught in Finnish from 03.09.2026–17.10.2026.

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1 Course information

The official course information is available on the University of Helsinki course page.

2 Lecture notes

2.1 03.09.2026

Definition 2.1. The set of *propositional formulas* is defined inductively as follows:

1. Every propositional symbol p_n , where $n \in \mathbb{N}$, is a propositional formula.
2. If A and B are propositional formulas, then

$$\neg A, \quad (A \wedge B), \quad (A \vee B), \quad (A \rightarrow B), \quad (A \leftrightarrow B)$$

are propositional formulas.

3. No other strings are propositional formulas.

Definition 2.2 (Immediate subformulas). The immediate subformulas of a propositional formula are defined by its outermost connective:

1. A propositional symbol p_n has no immediate subformulas.
2. The only immediate subformula of $\neg A$ is A .

3. The immediate subformulas of $(A \circ B)$ are A and B , where

$$\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}.$$

Definition 2.3 (Subformulas). A formula B is a *subformula* of a formula A if $B = A$, or if there are formulas

$$A = A_0, A_1, \dots, A_k = B$$

such that $k \geq 1$ and A_{i+1} is an immediate subformula of A_i for every $0 \leq i < k$.

Theorem 2.4 (Structural induction). *Let $P(A)$ be a statement about propositional formulas. Suppose that*

1. $P(p_n)$ holds for every $n \in \mathbb{N}$;
2. if $P(A)$ holds, then $P(\neg A)$ holds; and
3. if $P(A)$ and $P(B)$ hold, then $P((A \circ B))$ holds for every

$$\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}.$$

Then $P(A)$ holds for every propositional formula A .

Example 2.5 (Structural induction). For a formula A , let $v(A)$ be the number of occurrences of propositional symbols in A , and let $b(A)$ be the number of occurrences of binary connectives in A . We prove that

$$v(A) = b(A) + 1$$

for every propositional formula A .

If $A = p_n$, then $v(A) = 1$ and $b(A) = 0$, so the claim holds. Suppose next that the claim holds for a formula A . Negation introduces neither a propositional symbol nor a binary connective, and hence

$$v(\neg A) = v(A) = b(A) + 1 = b(\neg A) + 1.$$

Finally, suppose that the claim holds for formulas A and B , and let $\circ$ be any binary connective. Then

$$\begin{aligned} v((A \circ B)) &= v(A) + v(B) \\ &= (b(A) + 1) + (b(B) + 1) \\ &= b((A \circ B)) + 1. \end{aligned}$$

The result follows by structural induction. □